1. (a)	$\left(1-\frac{1}{6}x\right)^9 = 1+9\left(-\frac{1}{6}x\right)^1+\frac{9\times 8}{2}\left(-\frac{1}{6}x\right)^2+\frac{9\times 8\times 7}{3!}\left(-\frac{1}{6}x\right)^3+\ldots$	M1
	$=1-\frac{3}{2}x+x^2-\frac{7}{18}x^3$	A1, A1
		(3)
(b)	$(10x+3)\left(1-\frac{3}{2}x+x^2-\frac{7}{18}x^3\right)$	
	Coefficient of $x^3 = 10\times 1+3\times -\frac{7}{18}=\frac{53}{6}$	M1 A1
		(2)
		(5 marks)

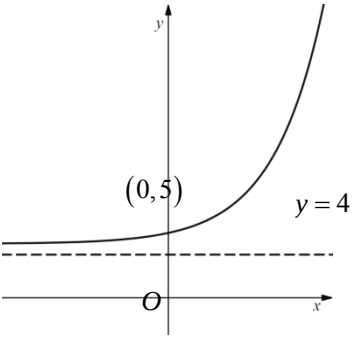
2.(a)	$a+5d=2$ and $\frac{10}{2}\{2a+9d\}=-80$	B1 B1
	Solves simultaneously $\Rightarrow d=20, a=-98$	M1 A1
		(4)
(b)	Attempts $\frac{n}{2}\{2\times-98+(n-1)\times 20\}...8000$	M1
	$5n^2-54n-4000...0\Rightarrow n=(34.2)$	dM1
	$n=35$	A1
		(3)
		(7 marks)

3 (i)	$2\log_2(2-x) = 4 + \log_2(x+10)$	
	One correct log law $2\log_2(2-x) \rightarrow \log_2(2-x)^2$ or $4 \rightarrow \log_2 16$	B1
	Correct attempt to combine two terms	M1
	Correct equation (not involving logs) E.g. $(2-x)^2 = 16(x+10)$	A1
	$x^2 - 20x - 156 = 0$ $(x+6)(x-26) = 0$	M1
	$x = -6$ only	A1
		(5)
(ii)	12	B1
		(1)
		(6 marks)

4.(a)	21	B1 **(1)**
(b)	Sets $f\left(\pm\frac{1}{2}\right)=0 \rightarrow$ equation in k	M1
	Eg. $\left(\frac{1}{2}-2\right)\left(2\times\frac{1}{4}+5\times\frac{1}{2}+k\right)+21=0$	
	$\Rightarrow 3+k=14 \Rightarrow k=11$ *	A1* **(2)**
(c)	(i) $f(x)=(x-2)(2x^2+5x+11)+21=2x^3+x^2+x-1$	M1
	$=(2x-1)(x^2+x+1)$	dM1, A1
	(ii) Attempts to find the number of roots of their (x^2+x+1)	M1
	• States (x^2+x+1) has no roots with reason	A1
	• Concludes $f(x)=0$ has 1 root (at $x=\frac{1}{2}$)	**(5)** **(8 marks)**

5 (a)	$(x-y)^3 = x^3 - 3x^2y + 3xy^2 - y^3$	M1
	$(x-y)^3 > x^3 - y^3 \Rightarrow -3x^2y + 3xy^2 > 0$	A1
	$\Rightarrow 3xy(y-x) > 0$	dM1
	As x and y are positive numbers $(y-x) > 0 \Rightarrow y > x$ *	A1*
		(4)
(b)	Chooses a suitable counter example. E.g. $x = 3, y = -1$	M1
	Shows that $(3--1)^3 > (3)^3 - (-1)^3$ as $64 > 28$ (BUT $-1 < 3$)	A1
		(2)
		(6 marks)

6.(a)



Shape **or** asymptote M1
Intercept B1
Fully correct A1

(3)

(b) $h = 0.3$ B1

$$\text{Area} \approx \frac{0.3}{2}\left\{0 + 4.3137 + 2\times\left(0.3246 + 0.8629 + 1.6643 + 2.7896\right)\right\}$$ M1

$$= \text{awrt } 2.34$$ A1

(3)

(c) (i) $$\int_2^{3.5}\left(2^x + 2x\right)\mathrm{d}x = \int_2^{3.5}\left(2^x - 2x + 4x\right)\mathrm{d}x = 2.34 + \left[2x^2\right]_2^{3.5}$$ M1

$$= 2.34 + 16.5 = 18.84$$ A1ft

(ii) $$\int_2^{3.5}\left(2^{x+1} - 4x\right)\mathrm{d}x = \int_2^{3.5} 2\left(2^x - 2x\right)\mathrm{d}x = 2\times 2.34 = 4.68$$ B1 ft

(3)

(9 marks)

7 (a)	$x^2 + y^2 + 8x - 10y = 29$	
	Attempts $(x+4)^2 + (y-5)^2 \ldots = \ldots$	M1
	Correct centre $(-4, 5)$	A1
	Exact radius $\sqrt{70}$	A1
		(3)
(b)	Attempts distance (or distance ^{2}) from "$(-4, 5)$" to "$(5, -8)$"	M1
	Full method. E.g. compares $\sqrt{9^2 + 13^2}$ to $\left(\sqrt{70} + \sqrt{52}\right)$	dM1
	States that • distance between centres $= \sqrt{250} = 15.81$ • sum of radii $= \sqrt{70} + \sqrt{52} = 15.58$ Concludes as sum of radii < distance between centres, result proven *	A1*
		(3) **(6 marks)**

8 (i)	States or uses $\tan x = \dfrac{\sin x}{\cos x} \Rightarrow 5\sin x \times \dfrac{\sin x}{\cos x} + 13 = \cos x$	B1
	$5\sin^2 x + 13\cos x = \cos^2 x \Rightarrow 5\left(1-\cos^2 x\right) + 13\cos x = \cos^2 x$	M1
	$\Rightarrow 6\cos^2 x - 13\cos x - 5 = 0$	A1
	$\Rightarrow (3\cos x + 1)(2\cos x - 5) = 0 \Rightarrow \cos x = -\dfrac{1}{3}$	M1
	$\Rightarrow x = 1.91$	A1
		(5)
(ii) (a)	$20 = 10 + 12\sin(6k+18)^\circ \Rightarrow \sin(6k+18)^\circ = \dfrac{5}{6}$	M1
	$\Rightarrow (6k+18) = 56.4, 123.6$	dM1
	$\Rightarrow k = 6.41, 17.59$	A1, A1
		(4)
(b)	22°C	B1
		(1)
(c)	$"6.41"t + 18 = 90 \Rightarrow t = 11.23$ Time of day = 11:14	M1, A1
		(2)
		(12 marks)

9.(a)	$y = 2x^{\frac{3}{2}}(4-x) = 8x^{\frac{3}{2}} - 2x^{\frac{5}{2}}$	
	$\frac{dy}{dx} = 12x^{\frac{1}{2}} - 5x^{\frac{3}{2}}$	M1, A1
	Stationary point $12x^{\frac{1}{2}} - 5x^{\frac{3}{2}} = 0 \Rightarrow x = \frac{12}{5}$	dM1, A1
		(4)
(b)	$\int 8x^{\frac{3}{2}} - 2x^{\frac{5}{2}} dx = \frac{16}{5}x^{\frac{5}{2}} - \frac{4}{7}x^{\frac{7}{2}}$	M1, A1
	$\frac{16}{5}k^{\frac{5}{2}} - \frac{4}{7}k^{\frac{7}{2}} = 0 \Rightarrow k = \frac{16\times 7}{4\times 5} = \frac{28}{5}$	dM1, A1
		(4)
		(8 marks)

10 (a)	$2000\times1.03^{6} = \text{awrt } 2390$	M1, A1 **(2)**
(b)	$3690 = ab^{4}, 3470 = ab^{7} \Rightarrow b^{3} = \dfrac{3470}{3690} = (0.94...)$	M1
	Hence $b = \sqrt[3]{\dfrac{3470}{3690}} = 0.9797...$ $a = \dfrac{3690}{0.9797..^{4}}$	dM1
	$N = 4000\times0.98^{t}$	A1 **(3)**
(c)	Sets $2000\times1.03^{T} = 4000\times0.98^{T} \Rightarrow \left(\dfrac{1.03}{0.98}\right)^{T} = \dfrac{4000}{2000}$	M1
	$\Rightarrow T\log\left(\dfrac{1.03}{0.98}\right) = \log 2 \Rightarrow T = ...$	dM1
	$\Rightarrow T = 13.9$	A1 **(3)** **(8 marks)**